



Modelling Financial Markets

Seminar 16: Digital Assets and DeFi — Hands-on



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Acronyms used in this material

- ▣ AI: Artificial Intelligence
- ▣ AR: AutoRegressive (model)
- ▣ CI: Confidence Interval
- ▣ ETF: Exchange-Traded Fund
- ▣ ETHA: iShares Ethereum Trust ETF (ticker)
- ▣ LP: Liquidity Provider
- ▣ NAV: Net Asset Value
- ▣ SE: Standard Error
- ▣ US: United States
- ▣ USDC: USD Coin (stablecoin issued by Circle)

Outline

Introduction

Part A: Pen and Paper

Part B: Estimation and Inference

Part C: Open Question and AI Critique

Wrap-up

References

Seminar Objectives

By the end of this seminar you can

- ▣ Compute a market-value weighted crypto index and its divisor step by step
- ▣ Price a swap in a constant-product pool and measure impermanent loss
- ▣ Explain stablecoin and ETF pegs as arbitrage bands
- ▣ Test crypto facts with block-bootstrap intervals and robust standard errors

- ▣ Background: Lecture 16; textbooks [Franke, Härdle & Hafner \(2019\)](#)

Structure

- ▣ **Part A:** eight pen-and-paper exercises (A1–A8)
- ▣ **Part B:** eight computational exercises with inference (B1–B8)
- ▣ **Part C:** an open question, a seed for the team project (C1), and an AI answer to critique (C2)

How to Work in This Seminar

- **[Solved]** problems show the full worked solution; **[Proposed]** problems show only the task: their solutions are discussed in class
- Every proposed problem names its **model**: the solved problem that uses the same technique
 - ▶ Every Part B exercise ends with interpretation questions
- Crypto trades 7 days a week: annualise with 365 days; joint analyses with shares use common days (prices joined first)
- Report every estimate with its uncertainty: bootstrap interval or robust standard error

Solved and Proposed Problems

Solved: worked examples in this deck

- Part A: A1 (index and divisor), A3 (a swap in a pool), A5 (impermanent loss), A7 (stablecoin arbitrage)
- Part B: B1 (Bitcoin volatility and the weekend effect), B3 (USD Coin peg dynamics), B5 (IBIT tracking), B7 (Strategy's beta)

Proposed: you solve them

- Part A: A2 (replacing a constituent; model: A1), A4 (arbitrage to the market price; model: A3), A6 (fees against losses; model: A5), A8 (ETF premium; model: A7)
- Part B: B2 (Ether and Solana; model: B1), B4 (Tether and Dai; model: B3), B6 (ETHA; model: B5), B8 (Coinbase; model: B7)
- Part C: C1, open question and project seed; C2, critique of an AI answer (Sharpe ratio and stablecoins)

Setup: Python Environment and Data

-  **Open in Colab:** `chapter16_seminar_notebook.ipynb`
- Google Colab (recommended) or local Python ≥ 3.10 with `numpy`, `pandas`, `scipy`, `statsmodels`, `matplotlib`
- Series: Bitcoin, Ether, Solana; Tether, USD Coin, Dai; IBIT, ETHA, Coinbase, Strategy; S&P 500, Nasdaq 100 (QQQ), gold (GLD), long Treasuries (TLT); daily closes up to 18 September 2026
- Crypto: 7 days a week; shares and ETFs: weekdays, prices adjusted for dividends and splits
- Weekly returns: Friday closes on common days, then log returns

Formulas Used Today

- ▣ Laspeyres index: $I_t = \sum_i P_{i,t} Q_{i,\tau} / D_\tau$; at a reallocation $D_{\tau+1} = \sum_i P_{i,t} Q_{i,\tau+1} / I_t$
- ▣ Constant product $xy = k$, fee f , $\gamma = 1 - f$: $\Delta y = y\gamma\Delta x / (x + \gamma\Delta x)$ Adams, Zinsmeister & Robinson (2020)
- ▣ Impermanent loss $IL(r) = 2\sqrt{r}/(1+r) - 1$; loss versus rebalancing $\sigma^2/8$ a year Milionis et al. (2022)
- ▣ Arbitrage band of a redeemable token: $1 - c_r \leq P \leq 1 + c_m$ Lyons & Viswanath-Natraj (2023)
- ▣ AR(1) half-life $h = \ln 0.5 / \ln \phi$; moving-block bootstrap with blocks of 20–21 days Künsch (1989)
- ▣ Hedge and safe haven Baur & Lucey (2010): $r_t^{\text{BTC}} = a + b r_t^{\text{S\&P}} + \sum_q c_q r_t^{\text{S\&P}} \mathbf{1}\{r_t^{\text{S\&P}} \leq q\} + u_t$

A1: A Three-Coin Index and Its Divisor [Solved]

Task

- Base day: prices $P_0 = (60\,000; 3\,000; 150)$ USD, supplies $Q = (19.5; 120; 500)$ million for coins A, B, C
- (a) Market values, weights and the divisor D_0 that sets $I_0 = 1000$
- (b) Month end: $P_1 = (66\,000; 2\,700; 180)$: the index level I_1 ; check with the weights
- (c) Reallocation: new supplies (19.52; 120.1; 560) million: the new divisor and weights

Solution

- Market values 1 170, 360, 75 billion (total 1 605);
 $D_0 = 1\,605 \cdot 10^9 / 1000 = 1.605 \cdot 10^9$
- Weights: 72.9%, 22.4%, 4.7%
- $I_1 = \sum_i P_{i,1} Q_i / D_0 = 1059.81$; check:
 $72.9\% \cdot 10\% - 22.4\% \cdot 10\% + 4.7\% \cdot 20\% = 5.98\%$
- $D_1 = \sum_i P_{i,1} Q_i^{\text{new}} / I_1 = 1.6167 \cdot 10^9$; new weights 75.2%, 18.9%, 5.9%
- The level stays 1059.81 on the switch day; only future returns use the new weights

A2: Replacing a Constituent [Proposed]

Solution discussed in class

Task

- After A1(c), coin C is delisted and coin D enters: $P_D = 0.5$ USD, $Q_D = 40$ billion coins
- (a) Market value of the old and the new basket on the switch day
- (b) The new divisor that keeps I continuous
- (c) Why does the divisor fall, and what would happen without the adjustment?
- Model: A1

A3: A Swap in a Constant-Product Pool [Solved]

Task

- Pool: $x = 1\,000$ Ether, $y = 3\,000\,000$ USD Coin, fee $f = 0.3\%$ Adams, Zinsmeister & Robinson (2020)
- (a) USD Coin received for selling 50 Ether and the execution price
- (b) Shortfall against the pool price 3 000 and the part due to the fee
- (c) The new pool price

Solution

- $\gamma \Delta x = 0.997 \cdot 50 = 49.85$;
$$\Delta y = \frac{3\,000\,000 \cdot 49.85}{1\,000 + 49.85} = 142\,449 \text{ USD Coin}$$
- Execution price $142\,449/50 = 2848.98$; shortfall $1 - 2848.98/3\,000 = 5.03\%$
- Without the fee: $\Delta y = 142\,857$, so the fee costs 408.22 USD Coin; the rest is price impact ($\delta = 5\%$)
- New price $(3\,000\,000 - \Delta y)/1\,050 = 2721.48$, i.e. -9.28%

A4: Arbitrage Back to the Market Price [Proposed]

Solution discussed in class

Task

- Same pool as A3 before the swap; the Ether price on other exchanges rises to 3 300 USD; ignore the fee
- (a) Reserves x_1 , y_1 when the pool price equals 3 300 (use $xy = k$, $y/x = p$)
- (b) USD Coin paid in, Ether taken out, average price and the arbitrageur's profit
- (c) Who pays the profit?
- Model: A3

A5: Impermanent Loss, Step by Step [Solved]

Task

- A liquidity provider deposits 10 000 USD in Ether and 10 000 USD in USD Coin at price p_0
- (a) Show that the position is worth $V_0\sqrt{r}$ when the price becomes $p_1 = rp_0$
- (b) Value of holding the two tokens and the impermanent loss for $r = 2$
- (c) IL for $r = 0.5, 1.5$ and 4

Solution

- $x_1 = \sqrt{k/p_1}, y_1 = \sqrt{kp_1},$
 $V_1 = p_1x_1 + y_1 = 2\sqrt{kp_1} = 2\sqrt{kp_0}\sqrt{r} = V_0\sqrt{r}$
- $r = 2$: LP $20\,000\sqrt{2} = 28\,284$; holding
 $10\,000 \cdot 2 + 10\,000 = 30\,000$
- $IL(2) = 2\sqrt{2}/3 - 1 = -5.72\%$ (a shortfall of 1 716 USD)
- $IL(0.5) = -5.72\%, IL(1.5) = -2.02\%, IL(4) = -20.00\%$:
 symmetric in $\ln r$, small for small moves

A6: Can Fees Pay for the Loss? [Proposed]

Solution discussed in class

Task

- The price in A5 doubles over one year
- (a) Fee income, in % of the position, that makes the provider as rich as holding
- (b) With volatility $\sigma = 70\%$, the loss versus rebalancing $\sigma^2/8$ a year [Milionis et al. \(2022\)](#)
- (c) With a 0.3% fee, the daily volume, as a share of the pool, that covers (b)
- Model: A5

A7: Stablecoin Redemption Arbitrage [Solved]

Task

- A stablecoin trades at 0.97 USD; the issuer redeems at 1 USD minus a 0.1% fee; a fixed transfer cost of 25 USD per operation
- (a) Profit from buying 1 million coins and redeeming them
- (b) The lower edge of the arbitrage band for a 1 million trade
- (c) Why did the band fail for USD Coin on 11 March 2023?

Solution

- Buy: 970 000 USD; redeem: 999 000 USD; profit $999\,000 - 970\,000 - 25 = 28\,975$ USD (2.99%)
- Break-even price: $(999\,000 - 25)/10^6 = 0.9990$; the band's lower edge is about $1 - c_r = 0.999$
- On a Saturday with a failed bank holding reserves, redemption could not settle and its value was uncertain: the arbitrage was no longer riskless [Diop, Chevallier & Sanhaji \(2024\)](#)

A8: The Premium of a Spot Bitcoin ETF [Proposed]

Solution discussed in class

Task

- ▣ Assume a fund held 0.000568 Bitcoin per share at launch and charges a 0.25% annual fee paid in Bitcoin [BlackRock \(2026\)](#)
- ▣ (a) Bitcoin per share after 2.5 years and the net asset value (NAV) with Bitcoin at 95 000 USD
- ▣ (b) The share trades at 53.10 USD: premium or discount?
- ▣ (c) With creation and redemption costs of 0.1%, what does an authorised participant do?
- ▣ Model: A7

B1: Bitcoin's Volatility and the Weekend Effect [Solved]

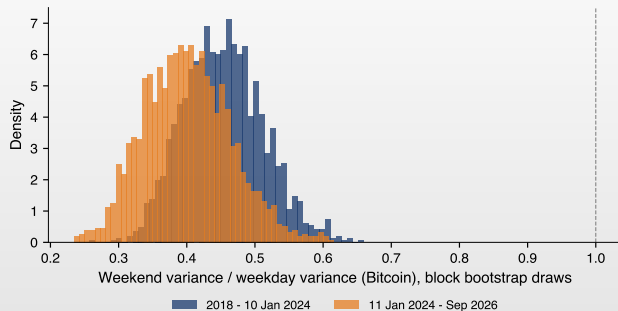
Task

- ▣ Bitcoin daily log returns, 2018 – September 2026 (3 182 days); S&P 500 on its own calendar
- ▣ (a) Annualised volatility (365 days) with a 95% moving-block bootstrap interval (blocks of 20 days, 2 000 resamples) [Künsch \(1989\)](#)
- ▣ (b) Weekend variance / weekday variance before (to 10 January 2024) and after the spot ETFs, each with a block-bootstrap interval
- ▣ (c) An interval for the change in the ratio

Interpretation

- ▣ Why blocks and not single days?
- ▣ Is the weekend lull a new, ETF-driven effect?
- ▣ What would a ratio of 1 mean?

B1: The Weekend Ratio Before and After the ETFs [Solved]



- Block-bootstrap distributions of the weekend / weekday variance ratio; the dashed line marks a ratio of 1

B1: Results [Solved]

- ▣ Volatility 64.1%, 95% CI [58.2, 70.4]; S&P 500 19.2%: ratio 3.3; excess kurtosis 14.9
- ▣ Weekend ratio: before 0.45 [0.35, 0.57] (2 200 days), after 0.39 [0.29, 0.53] (982 days)
- ▣ Change -0.06 , 95% CI $[-0.23, 0.11]$: contains 0, so no evidence that the ETFs changed the weekend effect
- ▣ Blocks keep volatility clusters intact; single-day resampling would give intervals that are too narrow
- ▣ Both ratios are far below 1: weekend returns have less than half the variance of weekday returns, before and after 2024

B2: Ether and Solana [Proposed]

Solution discussed in class

Task

- Repeat B1 for Ether (2018–2026) and Solana (April 2020 – 2026); compare the volatility with the S&P 500 on the same period
- (a) Volatility with a block-bootstrap interval; (b) weekend ratio before and after the ETFs and the change
- Model: B1

Interpretation

- Which coin is the riskiest relative to shares?
- Is the weekend effect weaker for coins with more on-chain activity?

B3: How Fast Does USD Coin Return to Par? [Solved]

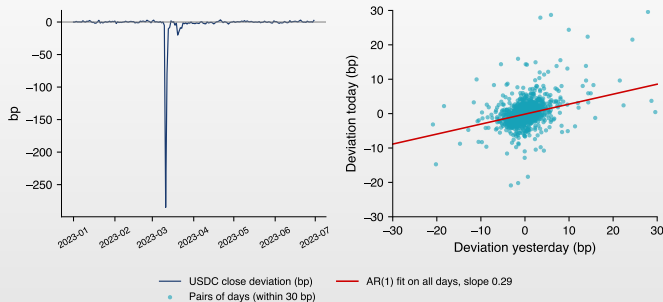
Task

- ▣ Daily close deviation $d_t = 10^4(P_t - 1)$ in basis points, 2021–2026 (2 087 days)
- ▣ (a) Standard deviation, days with $|d_t| > 50$ bp, lowest close and lowest daily low
- ▣ (b) AR(1) fit $d_t = c + \phi d_{t-1} + \varepsilon_t$, half-life, and a 95% residual-bootstrap interval for ϕ
- ▣ (c) Repeat without 9–16 March 2023

Interpretation

- ▣ Does one week drive the statistics?
- ▣ Is a Normal-distribution VaR of d_t meaningful?
- ▣ What does a half-life below one day say about arbitrage?

B3: USD Coin Deviations [Solved]



- Left: close deviation in the first half of 2023; right: today's against yesterday's deviation, zoomed to ± 30 bp, with the AR(1) line fitted on all days

B3: Results [Solved]

- s.d. 7.5 bp (without March 2023: 3.7); 3 days beyond 50 bp (without: 1); lowest close −285 bp, lowest low −1226 bp (11 March 2023)
- $\hat{\phi} = 0.290$ [0.257, 0.317], half-life 0.56 days [0.51, 0.60]; without March 2023: $\hat{\phi} = 0.308$, half-life 0.59
- One week doubles the standard deviation but barely moves $\hat{\phi}$: the dynamics are robust, the scale is not
- Excess kurtosis 1031: a Normal VaR would be useless; the risk is a rare jump, not daily noise
- Half of a deviation is gone within a day: arbitrage through minting and redemption is fast in normal times Lyons & Viswanath-Natraj (2023)

B4: Tether and Dai [Proposed]

Solution discussed in class

Task

- Repeat B3 for Tether and Dai, 2021–2026
- (a) Deviation statistics; (b) AR(1) with bootstrap interval and half-life, with and without March 2023
- Model: B3

Interpretation

- Which coin returns to par most slowly, and why might that be?
- Which coin had its worst day outside March 2023?

B5: How Well Does IBIT Track Bitcoin? [Solved]

Task

- ▣ IBIT and Bitcoin on IBIT's trading days since 11 January 2024 (673 days, 140 weeks)
- ▣ (a) Regress IBIT on Bitcoin returns, daily and weekly (Friday to Friday), with Newey–West standard errors; test $\beta = 1$ weekly
- ▣ (b) Tracking error, daily and weekly
- ▣ (c) Slope of $\ln(P^{\text{IBIT}}/P^{\text{BTC}})$ on time, with a Newey–West 95% CI; compare with the 0.25% fee [BlackRock \(2026\)](#)

Interpretation

- ▣ Why is the daily R^2 lower?
- ▣ Is the fee visible in the data?
- ▣ Why use Newey–West errors for (c)?

B5: Results [Solved]

- Daily: $\hat{\beta} = 0.93$, $R^2 = 0.82$; weekly: $\hat{\beta} = 1.007$ (SE 0.011, $t = 0.63$ for $\beta = 1$), $R^2 = 0.98$
- Tracking error: 20.8% (daily) vs 6.2% (weekly)
- The daily gap comes from the two closing times; the next day's Bitcoin return adds almost nothing ($\hat{\beta}_{t+1} = 0.04$)
- Drift -0.237% a year, 95% CI $[-0.315, -0.158]$: contains the 0.25% fee, excludes 0
- The residuals of a regression of a log ratio on time are strongly autocorrelated: ordinary standard errors would be far too small

B6: ETHA and Ether [Proposed]

Solution discussed in class

Task

- Repeat B5 for ETHA (the spot Ether ETF, from 23 July 2024) against Ether
- Model: B5

Interpretation

- Is ETHA a worse tracker than IBIT, or only noisier?
- How many years of data would pin the fee down to $\pm 0.05\%$?

B7: Is Strategy a Leveraged Bitcoin? [Solved]

Task

- Weekly returns of Strategy (formerly MicroStrategy), Bitcoin and the S&P 500, April 2021 – September 2026
- (a) Two-factor regression with Newey–West errors; test $\beta_{BTC} = 1$
- (b) The same on April 2021 – 2023 and on the ETF period

Interpretation

- Is the Bitcoin beta stable?
- Is the annualised $\hat{a}$ (alpha) evidence of skill?

B7: Results [Solved]

	n	$\hat{\beta}_{\text{BTC}}$ (SE)	$t(\beta_{\text{BTC}} = 1)$	$\hat{\beta}_{\text{SPX}}$ (SE)	R^2	alpha, % a year (t)
Apr 2021 – Sep 2026	284	1.18 (0.11)	1.64	1.02 (0.24)	0.63	−4.5 (−0.19)
Apr 2021 – Dec 2023	142	1.01 (0.14)	0.10	1.40 (0.35)	0.65	0.3 (0.01)
Jan 2024 – Sep 2026	141	1.50 (0.12)	4.12	0.48 (0.29)	0.65	−9.3 (−0.29)

- Before 2024 Strategy moved one for one with Bitcoin ($\hat{\beta}_{\text{BTC}} = 1.01$) plus a large equity beta
- Since 2024 the Bitcoin beta rose to 1.50 ($t = 4.12$ against 1): more sensitivity to Bitcoin, less to the stock market
- Alpha is never significant ($|t| < 1$): no evidence of skill beyond the factor exposures

B8: Coinbase [Proposed]

Solution discussed in class

Task

- Repeat B7 for Coinbase (listed from April 2021)
- Model: B7

Interpretation

- Is Coinbase a Bitcoin proxy or a US growth stock?
- Is the large negative alpha significant?

C1: Alternative Asset, Risk Asset or Hedge?

Question

- Is Bitcoin an alternative asset, a risk asset or a hedge? Evidence since the spot ETFs (11 January 2024) against January 2021 – January 2024
- Assets: S&P 500, Nasdaq 100 (QQQ), gold (GLD), long Treasuries (TLT); daily returns on common days
- Definitions of hedge and safe haven: [Baur & Lucey \(2010\)](#); earlier evidence for Bitcoin: [Bouri et al. \(2017\)](#)

What a good answer contains

- Correlations before and after, with block-bootstrap intervals for the change
- A hedge / safe-haven regression with extreme-quantile interactions and robust errors
- Downside vs upside beta; Bitcoin on the ten worst S&P 500 days
- A link to the distribution features of the lecture ([Pele et al. \(2023\)](#)): “alternative” in shape, “risk” in co-movement?

C2: Critique an AI Answer [Proposed]

- A student asks an AI assistant: “Compare Bitcoin with lending USD Coin (USDC) at 5% a year: compute Bitcoin’s annualised Sharpe ratio on daily data for 2021–2025, with the USDC yield as the risk-free rate”
- The assistant answers with this code:

```
import numpy as np, pandas as pd
btc = pd.read_csv('BTC-USD.csv', parse_dates=['date']).set_index('date')['close']
r = np.log(btc.loc['2021':'2025']).diff().dropna()    # daily log returns
mu = r.mean() * 252                                # annual mean
sd = r.std() * np.sqrt(252)                          # annual volatility
rf = 0.05                                           # USDC lending: risk-free
sharpe = (mu - rf) / sd
print(f"mu={mu:.1%}, vol={sd:.1%}, Sharpe={sharpe:.2f}")
```

- “Output: mu = 15.1%, vol = 48.5%, Sharpe = 0.21. USDC is fully backed by dollars and has never traded below \$0.99, so lending it is a risk-free investment”

C2: Critique an AI Answer, Task [Proposed]

Solution discussed in class

Task

- The answer contains **three** errors, in the code and in the text; everything else is correct
- For each error
 - ▶ say what is wrong and why
 - ▶ say how you would detect it: a count in the data, the data themselves, a definition from the lecture
 - ▶ give the corrected line or statement
- Rerun the corrected code; report the mean, the volatility and the Sharpe ratio
- Log the three errors as in `AI_ERRORS.md`: what was asked, what the AI answered, what was wrong, how you found out, the correct version

Interpretation

- How many daily returns should 2021–2025 contain for Bitcoin? And for the S&P 500?
- Which of the three errors can you check in one line with the course data?
- If lending USDC were risk-free, why would it pay more than a Treasury bill?

Common Pitfalls

- ▣ Forgetting the divisor when an index changes constituents or supplies
- ▣ Using the pool price as the execution price: large trades pay the price impact
- ▣ Judging a liquidity position against holding instead of against rebalancing
- ▣ Treating stablecoin deviations as Normal noise; they are rare jumps
- ▣ Daily regressions between assets that close at different times; ordinary standard errors on trending ratios
- ▣ Calling Bitcoin a hedge without testing the extreme-day betas

Take-Home Assignment (Team Project)

- ▣ Choose three crypto-assets not analysed in class (or a stablecoin and two coins)
- ▣ Build a market-value weighted index with a divisor and compare it with Bitcoin: return, volatility, tracking error, with bootstrap intervals
- ▣ Run the hedge / safe-haven regression of Part C for your index against the S&P 500 and gold
- ▣ Deliver a Quantlet folder (code, Metainfo, charts) and three slides: numbers, interpretation, limitations
- ▣ Log every AI error you catch in `AI_ERRORS.md`, as in C2: what was asked, what was wrong, how you found out, the correct version

Further Study on Quantinar

Advanced courses on quantinar.com

- ▣  Are Cryptos becoming alternative Assets?
- ▣  Comparison of Cryptocurrency Indices
- ▣  Stabilising Onchain Stablecoin Risks
- ▣  On Crypto-backed Loans

Code and data

- ▣ Seminar notebook (Colab); Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_16

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